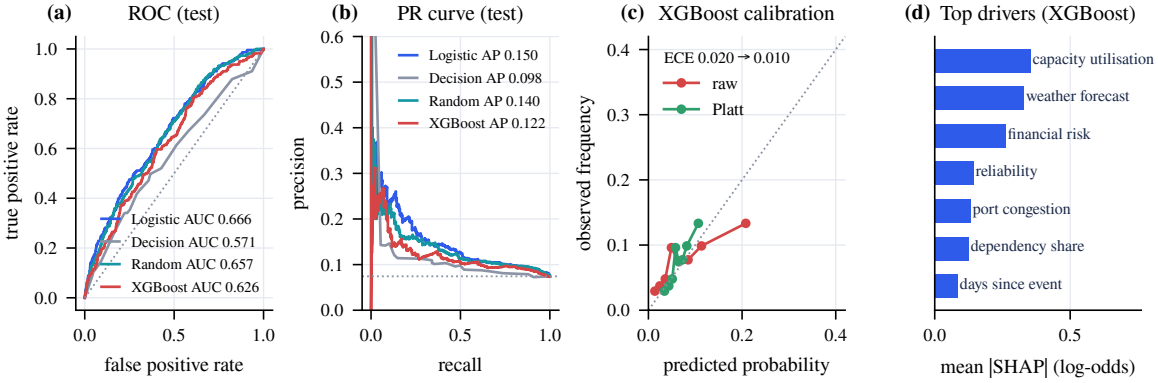
(a)
ROC (test)
true positive rate
false positive rate
Logistic AUC 0.666
Decision AUC 0.571
Random AUC 0.657
XGBoost AUC 0.626
(b)
PR curve (test)
precision
recall
Logistic AP 0.150
Decision AP 0.098
Random AP 0.140
XGBoost AP 0.122
(c)
XGBoost calibration
observed frequency
predicted probability
ECE 0.020 → 0.010
raw
Platt
(d)
Top drivers (XGBoost)
capacity utilisation
weather forecast
financial risk
reliability
port congestion
dependency share
days since event
mean |SHAP| (log-odds)